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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (CYBER SECURITY)

MICRO PROJECT REPORT

VibeGuard: An Embedded Vibration Condition Monitoring System

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in partial fulfillment for the award of the degree

of

BACHELOR OF TECHNOLOGY (B.Tech)

in

COMPUTER SCIENCE & ENGINEERING (CYBER SECURITY)

of

A P J ABDUL KALAM TECHNOLOGICAL UNIVERSITY

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SEPTEMBER 2026

DECLARATION

We hereby declare that the report entitled “VibeGuard: An Embedded Vibration Condition Monitoring System”, submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology of the A P J Abdul Kalam Technological University, Kerala, is a bona fide work carried out by us. The technical description in this report is based primarily on the controlled project records maintained for the VibeGuard semester project and on the cited external technical documentation. Wherever ideas, specifications, or statements from other sources have been used, they have been appropriately acknowledged and referenced. We further declare that no experimental result has been fabricated and that quantities not yet measured are identified as such.

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BONAFIDE CERTIFICATE

This is to certify that the project report entitled **VibeGuard: An Embedded Vibration Condition Monitoring System** submitted by **Nihad P C, Sreehari K, Amith Krishna Das, Sreeprada K S and Archa Pramod** to the A P J Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the **Bachelor of Technology** degree in **Computer Science & Engineering (Cyber Security)** is a record of the project work carried out under the guidance and supervision of the project supervisor.

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PROGRAMME EDUCATIONAL OBJECTIVES (PEOs)

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- PEO2.Graduates will be able to advance themselves and to adapt quickly to new environments and technologies, assimilate new information efficiently, and thrive in multidisciplinary settings with a strong focus on innovation and entrepreneurship.
- PEO3.Graduates will be to apply the knowledge of various tools and techniques gained through program to identify, counter and recover from cyber threats.

PROGRAMME SPECIFIC OBJECTIVES (PSOs)

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- PSO2. Graduates will gain knowledge with respect to ever growing threats including both hardware and software.
- PSO3.Graduates develop skills to anticipate future cyber threats without compromising Cyber laws and individual's privacy

Course Outcomes (COs)

- CO1:Identify the basic structure and functional units of a digital computer and the features of RISC architecture. (K2).
- CO2: Experiment with the single cycle processor, pipelining, and the associated problems.(K3)
- CO3: Utilize the memory organization in modern computer systems.(K3)
- CO4:Experiment with the I/O organization of a digital computer.(K3)

PO - CO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
PBCST 404.1	3	2	-	-	2	-	-	-	-	-	1	2	3	-
PBCST 404.2	3	3	1	2	3	-	-	1	-	-	1	3	3	1
PBCST 404.3	3	2	-	1	2	-	-	-	-	-	1	2	3	-
PBCST 404.4	2	2	1	2	3	-	-	1	1	-	1	2	3	1

ACKNOWLEDGEMENT

We take this opportunity to express our sincere gratitude to all those who guided, encouraged and supported us during the development of the VibeGuard project and the preparation of this report.

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We also acknowledge the project records and engineering documentation maintained by the VibeGuard team. The repository provided the controlled project architecture, component decisions, test procedures, evidence rules and safety boundaries that form the technical basis of this report.

Finally, we thank our teammates, friends and all teaching and non-teaching staff who contributed directly or indirectly to the successful planning and execution of the project.

ABSTRACT

VibeGuard is a small, local embedded vibration condition monitoring system designed to demonstrate how a low-cost sensor and microcontroller can transform vibration from a rotating test rig into an explainable abnormal-condition indication. The semester architecture uses one rigidly mounted ADXL345 three-axis accelerometer connected to an ESP32-family controller over SPI. The controller acquires timestamped samples, removes the static or DC component, calculates root-mean-square (RMS) vibration features, and applies a persistence rule so that a short disturbance does not immediately change the reported state. A triggered Hann-windowed Fast Fourier Transform (FFT) provides supporting frequency-domain evidence, while a common-cathode RGB LED and USB serial logging provide local state indication and traceable data output. The system is intentionally bounded to a guarded low-voltage 12 V motor or fan teaching rig and does not claim industrial fault diagnosis, cloud monitoring, active cancellation, permanent multi-sensor attribution, or remaining-useful-life prediction. The project repository defines a staged engineering process covering component acceptance, sensor readback, acquisition integrity, safe abnormal-condition generation, classifier verification, leakage-resistant dataset splitting, final configuration freeze and locked validation. At the current documented project state, quantitative performance results are not yet measured; therefore, this report distinguishes between verified architecture and proposed evaluation criteria and avoids presenting unmeasured values as results. The work demonstrates a practical embedded-systems methodology in which hardware identity, signal-processing decisions, experiment configuration and evidence provenance are treated as part of the technical system itself.

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LIST OF ABBREVIATIONS

ADC	: *Analog – to – DigitalConverter*
ADXL345	: *Three – AxisDigitalAccelerometer*
FFT	: *FastFourierTransform*
MCU	: *MicrocontrollerUnit*
ODR	: *OutputDataRate*
RMS	: *RootMeanSquare*
SPI	: *SerialPeripheralInterface*
USB	: *UniversalSerialBus*
DSP	: *DigitalSignalProcessing*
RUL	: *RemainingUsefulLife*

LIST OF NOTATIONS

x_i : i^{th} signal sample

N : $Number of samples in one analysis window$

$\bar{x}$: $Mean value of a window$

x_i' : $DC - removed sample$

f_s : $Sampling/output data rate$

f_k : $Frequency represented by FFT bink$

K, M : $Persistence parameters$

μ, Σ : $Mean vector and covariance matrix for an optional statistical comparator$

CHAPTER 1
INTRODUCTION

1.1 Overview

Rotating machines produce vibration as a natural consequence of motion, balance, mounting, load and structural interaction. When the operating condition changes, the vibration pattern can also change. The project repository identifies this observation as the starting point for VibeGuard, a small local vibration-condition monitor intended for a controlled teaching rig rather than for unrestricted industrial diagnosis. The system converts vibration into a measurable data stream using a rigidly mounted ADXL345 three-axis accelerometer and an ESP32-family controller. The controller acquires the measurements over SPI, performs local signal preprocessing, extracts vibration features and produces a local normal or abnormal indication. The project is deliberately narrow: its semester claim is that, under a documented configuration and test protocol, the system should be able to learn a normal vibration baseline and flag a safe, deliberately induced abnormal condition on a guarded low-voltage motor or fan rig [1].

The technical interest of VibeGuard lies not only in the sensing hardware but also in the discipline used to turn sensor data into a defensible engineering conclusion. The repository requires timestamped and counted acquisition blocks, explicit configuration identifiers, immutable raw data, versioned firmware and parameters, and leakage-resistant validation. The minimum decision baseline is RMS with persistence. A triggered Hann-windowed FFT is used as supporting frequency-domain evidence, while an ordinary Mahalanobis distance is explicitly optional and subordinate to the simpler baseline [1, 2]. This approach keeps the system explainable and compatible with a small embedded controller.

The project also treats physical safety and evidence management as part of the system design. The motor or fan rig is separately powered from the ESP32, the abnormal fixture must be positively retained and guarded, and adjustments to the rotating rig are made only with power disconnected. The repository further prohibits unsupported claims such as bearing-fault diagnosis, remaining-useful-life prediction, industrial performance certification or cloud dependence [1, 2]. The result is a project that combines embedded firmware, digital signal processing, experimental design, software configuration control and physical demonstration in one coherent workflow.

1.2 Objective

The primary objective of VibeGuard is to design and validate a compact embedded system that can distinguish a documented normal vibration condition from a deliberately introduced safe abnormal condition on a controlled motor or fan rig. The repository defines the following supporting objectives:

1. To acquire three-axis vibration data from an ADXL345 accelerometer using a documented SPI interface and preserve sample timing and integrity information.
2. To remove the static or DC component of each analysis window and calculate an explainable vibration magnitude feature using RMS.
3. To apply persistence logic so that a short disturbance does not immediately produce an abnormal state.
4. To add a triggered Hann-windowed FFT as supporting frequency-domain evidence without compromising acquisition integrity.
5. To provide a clear local state indication using a common-cathode RGB LED and struc-

tured USB serial logging.

6. To evaluate the design using session-level data separation, predefined metrics and a held-out validation procedure without tuning on test data.

7. To maintain traceability from hardware and firmware versions through parameters, datasets and recorded evidence.

1.3 Problem Statement

A person can sometimes notice a change in the sound or feel of a rotating mechanism, but subjective inspection is difficult to standardize. A teaching laboratory also needs a safe and repeatable way to demonstrate how a change in mechanical behaviour can be reflected in sensor data. The problem addressed by VibeGuard is therefore not to diagnose every possible machinery fault, but to build a bounded and explainable sensing-and-decision pipeline that can recognize whether the measured vibration remains consistent with a previously established normal baseline.

The problem has several engineering dimensions. First, the sensor stream must be acquired without unexplained sample loss. Second, the raw acceleration contains static components such as gravity and orientation offsets that should not dominate a vibration feature. Third, a single transient disturbance should not necessarily force an abnormal state. Fourth, any frequency analysis must be computationally bounded so that it does not interfere with sample capture. Finally, the evaluation must avoid data leakage: adjacent windows from the same physical run are not independent observations, so the test must be based on independent sessions, days or remounts [2].

1.4 Scope

The semester Level-1 scope contains one permanently used rigidly mounted ADXL345-class three-axis accelerometer, an ESP32-WROOM-32 development board or an explicitly authorized ESP32-S3 fallback, SPI acquisition, local preprocessing, RMS plus persistence, triggered FFT and selected bands, RGB indication, USB logging and a guarded low-voltage 12 V motor/fan test rig. The exact board, pinout, mount, rig, firmware, parameters and schema are identified and frozen as a complete configuration before locked validation [1, 2].

The scope explicitly excludes a permanent second accelerometer, dense source attribution, cloud dependence, active cancellation, industrial remaining-useful-life prediction, bearing-fault certification and the future PIRG multi-sensor research concept. These exclusions are important because they define what the project does and does not claim. The repository states that future research must preserve the semester baseline unchanged and use a separate configuration and protocol [1].

1.5 Organisation of the Report

The report is organised as follows:

- **Chapter 1: Introduction** presents the motivation, problem statement, objectives and scope of VibeGuard.
- **Chapter 2: Literature Survey** reviews vibration condition monitoring, MEMS accelerometers, embedded sensing, RMS and FFT-based analysis, and the importance of leakage-resistant validation.
- **Chapter 3: Methodology** explains the VibeGuard architecture, hardware selection, signal-

processing pipeline, decision logic, state handling, experiment design and evidence workflow.

- **Chapter 4: Results & Discussion** summarizes the project evidence currently available, the proposed evaluation criteria and the limitations that prevent unmeasured quantities from being reported as final results.
- **Chapter 5: Conclusion** summarizes the contribution of the project and identifies future directions that remain outside the current semester baseline.

CHAPTER 2
HISTORY

2.1 Evolution of Machine Condition Monitoring

Condition monitoring has developed from periodic manual inspection toward continuous measurement of physical quantities such as vibration, temperature, pressure and electrical current. In rotating machinery, vibration is particularly useful because changes in balance, mounting, loading and mechanical interaction can alter the dynamic response of the machine. Vibration-based condition monitoring therefore provides a way to observe a machine without requiring the monitored process to be dismantled. The engineering literature treats vibration as one of the important observables for condition assessment, while also emphasizing that interpretation depends on the machine, operating condition and measurement arrangement [7].

Traditional monitoring arrangements can use dedicated industrial accelerometers, signal-conditioning electronics, data acquisition equipment and desktop analysis software. These systems can provide high-quality measurements, but they may be unnecessarily complex for a small academic demonstration. A teaching project can instead focus on the complete reasoning chain: acquire a repeatable signal, remove known unwanted components, calculate an interpretable feature, apply a defined decision rule and preserve the evidence required to reproduce the result.

2.2 Development of MEMS-Based Vibration Sensing

The availability of low-cost microelectromechanical-system (MEMS) accelerometers has reduced the hardware required to demonstrate vibration measurement. The ADXL345 selected

for VibeGuard is a three-axis digital accelerometer supporting selectable ranges of ± 2, ± 4, ± 8 and ± 16 g, SPI or I^2C communication, and output data rates extending to 3200 Hz. Its operating supply range is specified as 2.0 to 3.6 V, making it compatible with common embedded controllers such as the ESP32 family [4].

MEMS sensing does not remove the need for careful experimental design. A sensor measures motion at its mounting point, so mounting stiffness, sensor orientation, cable movement and mechanical coupling can influence the recorded waveform. Consequently, VibeGuard treats the sensor mount as part of the configuration rather than as an incidental laboratory detail. The repository requires a documented mount identifier and treats a material change in mounting as a configuration change [2].

2.3 Rise of Embedded Local Processing

The next stage in the evolution of compact monitoring systems is the movement of signal processing closer to the sensor. A microcontroller can sample the sensor, execute lightweight numerical operations and communicate a decision without continuously sending raw data to a remote server. The ESP32-DEVKITC-32E family used as the preferred controller for VibeGuard exposes the required general-purpose I/O and supports the local execution model required by the project [5].

Local processing offers two educational benefits. First, it demonstrates how a real-time embedded system must balance acquisition timing with computation. Second, it makes the decision path observable: the project can log the acquisition state, computed features and output state from the same controller that performs the analysis. VibeGuard therefore treats cloud connectivity as unnecessary to the semester demonstration and keeps the primary decision path

local [1].

2.4 Development of Time-Domain Features

As embedded monitoring became more accessible, simple statistical features became attractive because they require fewer resources than large machine-learning pipelines. The root mean square (RMS) of a zero-mean vibration signal is one such feature. It summarizes signal magnitude over a window and can be calculated deterministically using a fixed number of arithmetic operations per sample. VibeGuard adopts RMS as the mandatory baseline because the repository requires the minimum viable decision method to remain explainable and verifiable [1].

The move toward simple features also highlights an important distinction between condition indication and fault diagnosis. A rise in RMS can indicate that the measured vibration has changed relative to the learned normal baseline, but it does not uniquely establish the physical cause of that change. The VibeGuard repository explicitly prohibits interpreting the red state as a bearing diagnosis or another specific industrial failure label [1].

2.5 Introduction of Frequency-Domain Analysis

Time-domain magnitude alone may not capture all useful information. Frequency-domain analysis can show where signal energy is concentrated and can help interpret changes that occur at particular frequency bands. The Fast Fourier Transform (FFT) became a practical way of computing discrete Fourier transforms efficiently, and embedded controllers can use short FFT windows when the implementation is computationally bounded.

VibeGuard adopts the FFT as supporting evidence rather than as the primary classifier. The

repository requires mean removal followed by Hann windowing before the FFT and specifies that the calculation should be triggered rather than performed on every acquisition window. This design is intended to reduce interference with the continuous acquisition path while still providing a frequency-domain view when the decision logic requests it [2].

2.6 Emergence of Reproducible Experimental Validation

Modern engineering projects increasingly distinguish between an algorithm that runs and an experiment that establishes evidence. In embedded vibration monitoring, adjacent windows collected from one run are correlated. Randomly dividing such windows between training and testing can therefore make a method appear more general than it is. The VibeGuard project adopts session-level separation so that windows from one physical session remain in the same split, with at least one held-out day or remount reserved for final evaluation [2].

This change in validation philosophy is important for a student project because it makes the final claim easier to audit. A result can be traced to a particular configuration, firmware version, parameter set, run and dataset instead of being represented only by a single accuracy value.

2.7 History of the VibeGuard Project

Within the project repository, VibeGuard was selected as the final semester project after an earlier multi-concept evaluation. The repository records Phase 3C as closed and Phase 4 as the active semester execution phase. The project scope was deliberately narrowed to one accelerometer, one ESP32-family controller, a guarded low-voltage motor or fan rig, local processing and a reproducible validation protocol [3, 1].

The current project history also shows a deliberate separation between the semester baseline and deferred research ideas. PIRG and other concepts are not part of the frozen VibeGuard MVP. This separation is intended to prevent the final demonstration from depending on features that have not undergone their own safety review, implementation verification or evidence plan [1].

2.8 Summary

The historical progression from manual observation to sensor-based condition monitoring, MEMS accelerometers and local embedded computation provides the technical context for VibeGuard. The project applies this progression in a deliberately constrained form: a small digital accelerometer, a microcontroller, explainable signal features and a controlled validation procedure. The emphasis is therefore not on reproducing the full complexity of industrial monitoring systems, but on demonstrating the engineering discipline required to build a measurement system whose claims remain proportional to its evidence.

CHAPTER 3
VIBEGUARD SYSTEM AND WORKING PRINCIPLE

3.1 System Architecture

VibeGuard is organized as a local measurement-and-decision pipeline. A guarded motor or fan rig provides the physical source of vibration. A rigidly mounted ADXL345 measures acceleration along three axes. The ESP32-family controller receives the digital samples using SPI, records timing and acquisition counters, performs preprocessing and calculates the decision features. The resulting state is displayed locally through an RGB LED and can be observed through structured USB serial records [1].

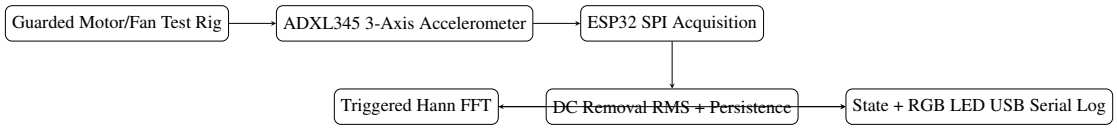


Figure 3.1: VibeGuard Level-1 local vibration-monitoring architecture

Figure 3.1 shows the principal signal path. The architecture deliberately separates continuous acquisition from optional spectral analysis. The FFT is triggered only when the decision logic requests additional frequency-domain evidence so that spectral processing does not become a hidden source of acquisition loss [2].

3.2 Hardware Components

Table 3.1: Principal VibeGuard hardware components

Component	Selected element	Purpose
Accelerometer	ADXL345	Three-axis digital vibration sensing
Controller	ESP32-WROOM-32 / authorized ESP32-S3 family	Sampling, processing, decision and logging
Interface	SPI	Deterministic digital sensor communication
Indicator	Common-cathode RGB LED	Local state indication
Test rig	Guarded 12 V DC motor/fan arrangement	Controlled normal and abnormal vibration generation
Power	Separate motor and ESP32 supplies	Electrical isolation and safe operation

The ADXL345 supports the measurement ranges and output data rates required for the project [4]. The repository's initial recommendation is full-resolution operation at ± 4 g with an 800 Hz output data rate and 256-sample analysis windows. These values are project parameters subject to pilot verification and final freeze rather than measurements obtained from completed validation [1].

3.3 Sensor Acquisition and Configuration

The preferred hardware mapping in the current project package assigns SPI clock to GPIO18, MOSI to GPIO23, MISO to GPIO19 and chip select to GPIO21. The ADXL345 interrupt line is assigned to GPIO4, while RGB channels are associated with GPIO25, GPIO26 and GPIO27. The repository identifies this as the proposed pin map and requires the as-built configuration to be confirmed before final freezing [1, 2].

During sensor bring-up, the playbook requires reading the device identity register, checking the expected ADXL345 identity value of 0xE5, reading back configuration registers and

performing stationary plausibility checks in multiple orientations. Six-axis-orientation testing is used as a simple diagnostic for sign, scaling and orientation mistakes before vibration experiments begin [2].

3.4 Signal Preprocessing

The raw acceleration includes a static component associated with gravity, orientation and sensor offsets. For an analysis window containing samples $x_1, x_2, \ldots, x_N$, the mean is calculated as

$$\bar{x} = \frac{1}{N} \sum_{i=1}^{N} x_i. \tag{3.1}$$

The zero-mean signal is then

$$x'_i = x_i - \bar{x}. \tag{3.2}$$

This step ensures that the vibration feature is not dominated by the static component. The same preprocessing definition must be used by embedded and offline analysis so that reference calculations remain comparable [2].

3.5 RMS and Persistence Decision Logic

The mandatory baseline feature is root mean square vibration magnitude:

$$\mathrm{RMS}(x') = \sqrt{\frac{1}{N} \sum_{i=1}^{N} (x'_i)^2}. \tag{3.3}$$

A threshold is not intended to be chosen after observing the final test results. The reposi-

tory instead requires pilot normal sessions to establish the baseline and then requires the threshold and persistence parameters to be frozen before locked testing. The persistence mechanism prevents one transient crossing from immediately changing the system to an abnormal state [2].

The project also defines an explicit Fault/Invalid state. Sensor identity failure, configuration mismatch, saturation, impossible values or acquisition overrun must not be silently interpreted as a normal vibration condition. This distinction improves the interpretability of the result because a missing or invalid measurement is represented as a data-quality state rather than as evidence of machine health [1].

3.6 Triggered Frequency-Domain Analysis

When the decision logic triggers spectral support, the selected window undergoes mean removal and Hann windowing before FFT processing. For a synthetic tone at a known frequency, the implementation should reproduce the expected spectral-bin location and normalization. The project playbook allows 256- or 512-point FFT configurations, with the final choice based on verification and resource measurement [2].

The FFT is not used as a complete fault classifier. It provides supporting frequency evidence, and dense source attribution is explicitly outside the semester scope. This distinction prevents the frequency plot from being interpreted as a diagnostic label when the available evidence supports only a condition-change indication.

3.7 Local State Machine and Output

The project defines the following conceptual states: Start/Self-check, Calibrating, Normal, Abnormal and Fault/Invalid. The RGB LED provides a simple human-readable indication

of these states. USB serial output provides the more detailed machine-readable record. The repository treats the LED as an interface to the state machine rather than as a diagnostic code for a specific physical fault [1].

The structured record is intended to preserve identifiers and status information such as run, configuration, firmware, parameters, dataset and schema references. This makes a later result traceable to the exact operating configuration instead of relying on an informal description written after the experiment.

3.8 Experimental Rig and Safety

The physical setup uses a guarded 12 V DC motor or fan arrangement. The motor supply is kept separate from the ESP32 USB supply, and the repository explicitly prohibits powering the motor from the ESP32 or its USB connection. The abnormal fixture must be positively retained and guarded, and physical changes to the rotating rig are performed only with power removed [1, 2].

3.9 Validation Methodology

The VibeGuard validation process is arranged as a sequence rather than as a single final test. Sensor bring-up establishes communication correctness. Deterministic vectors verify mathematical calculations. Controlled normal runs establish the baseline. Safe abnormal runs exercise the decision logic. Final validation is performed only after the hardware, firmware, parameter set and data schema are frozen.

The repository requires session-level splitting. All windows from a single physical session stay together, and at least one held-out day or remount is reserved for final evaluation. This

protects the final result from overly optimistic random-window splitting of highly correlated data [2].

Table 3.2: Repository-defined proposed locked-validation criteria

Metric	**Proposed criterion**
Separated sessions	At least 3 independent session days
Held-out evaluation	At least 1 held-out day or remount
Sensitivity	$\geq 90\%$
Specificity	$\geq 90\%$
False alerts	≤ 1 per 10-minute stable-normal run
Alert latency	≤ 2 s
Acquisition loss	Zero unexplained lost blocks
Reproducibility	Successful second-operator reproduction

These values are proposed acceptance criteria recorded by the project documents; they are not reported here as achieved performance. The separation is essential to maintain an accurate boundary between methodology and results.

3.10 Evidence and Configuration Control

The repository defines configuration identifiers in the form `CFG-###`, combining hardware revision, mount, rig, firmware commit, parameter set and schema. Raw data are intended to remain immutable, with supporting hashes and provenance records linking each final claim to the underlying run and configuration [2].

This evidence structure is especially useful for the VibeGuard project because mechanical mounting and firmware parameters can materially change the observed signal. A change to the physical system should therefore generate a new configuration identity rather than being silently mixed into an earlier dataset.

CHAPTER 4
ANALYSIS WITH OTHER APPROACHES

4.1 Purpose of the Comparison

VibeGuard is intended as a controlled embedded demonstration rather than a general-purpose industrial monitoring platform. The appropriate comparison is therefore with other broad approaches to vibration monitoring, focusing on architecture, computational burden, explainability and validation requirements rather than claiming that one approach is universally superior.

4.2 Manual Inspection

Manual inspection relies on an operator observing sound, touch, visible motion or other physical symptoms. It can be useful for quick checks, but the observation is subjective and difficult to quantify consistently across operators and sessions. VibeGuard replaces the central observation step with a digital measurement that can be logged and compared against a documented baseline.

4.3 Continuous Cloud-Based Monitoring

A cloud-based architecture can support remote dashboards, centralized storage and fleet-scale analytics. Those features are valuable in larger deployments, but they introduce communication, connectivity, account and data-management dependencies. The current VibeGuard architecture is intentionally local and does not require cloud connectivity for its primary decision path [1].

4.4 Full Spectrum Analysis on Every Window

One possible design is to compute a complete FFT for every analysis window. This provides continuous spectral information, but it also introduces a repeated computational cost. VibeGuard instead uses RMS and persistence as the minimum decision baseline and triggers FFT processing only when configured. The repository explicitly requires resource and acquisition-integrity measurements so that the spectral stage does not undermine sampling reliability [2].

4.5 Statistical Anomaly Detection

Statistical methods can combine multiple features and quantify distance from a learned normal distribution. The repository discusses ordinary Mahalanobis distance as an optional comparator. Its position in the design is deliberately secondary to the RMS baseline because the project should not depend on a more complex method unless the available data, schedule and validation evidence justify it [2].

4.6 Machine-Learning Classification

A supervised classifier could be trained on labelled normal and abnormal datasets. Such a system may capture nonlinear patterns that are difficult to express with a single threshold, but it also increases the data requirement, model-management burden and risk of leakage. VibeGuard avoids making machine learning a hidden dependency of the semester MVP. The repository instead defines a transparent baseline whose inputs and decisions can be explained directly.

4.7 Comparison of Architectural Characteristics

Table 4.1: VibeGuard compared with broad monitoring approaches

Approach	Processing	Primary output	Main validation issue
Manual inspection	Human	Subjective observation	Repeatability across operators
Cloud monitoring	Local + remote	Dashboard/alerts	Connectivity and data management
Full FFT pipeline	Embedded/high compute	Continuous spectrum	Resource and timing cost
Statistical anomaly detection	Embedded/software	Distance score	Adequate independent baseline data
Machine-learning classifier	Embedded/host	Class label/probability	Dataset size and leakage control
VibeGuard baseline	Local embedded	Normal/Abnormal + support data	Held-out session validation

4.8 Interpretation for the Semester Project

The comparison indicates that the main engineering decision in VibeGuard is not to reproduce every feature of a production monitoring platform. Instead, it is to establish a clear relationship between a modest sensor configuration, a bounded processing pipeline and an auditable experiment. The project can then demonstrate acquisition, preprocessing, feature extraction, state logic and validation as a connected embedded system.

The repository's exclusions are important in this context. There is no permanent second accelerometer, no dense source-attribution classifier, no cloud dependency and no remaining-useful-life prediction in the frozen semester baseline. Those exclusions are deliberate scope boundaries rather than missing requirements [1].

4.9 Summary

VibeGuard occupies a deliberately constrained point in the design space. It combines digital MEMS sensing, local embedded computation and explainable signal features while keeping the physical experiment controlled. The comparison does not establish a universal ranking among alternatives; instead, it explains why the chosen semester architecture emphasizes traceability, simplicity and reproducible validation.

CHAPTER 5
ADVANTAGES

5.1 Local and Explainable Decision Path

The main advantage of the VibeGuard architecture is the direct relationship between measurement and decision. The sensor produces acceleration values, the controller removes the static component, RMS quantifies the remaining vibration magnitude and persistence converts repeated threshold behaviour into a state decision. Each stage can be explained independently and verified with deterministic data [2].

5.2 Low Hardware Complexity

The semester baseline uses one three-axis accelerometer and one ESP32-family controller. A small number of components reduces wiring complexity and makes the test setup easier to inspect. The ADXL345 also provides digital SPI output, which removes the need for a separate analog amplifier and ADC chain for the sensor signal [4].

5.3 Real-Time Local Operation

The controller is responsible for acquisition, processing and local indication. This enables the project to demonstrate an embedded real-time pipeline without requiring an external computer for the primary state decision. USB serial logging is retained as an evidence and observation channel rather than as a dependency for the decision itself [1].

5.4 Use of Multiple Axes

Although the minimum classifier can be based on magnitude features, the three-axis sensor provides additional information about the vibration direction. The project can therefore inspect individual axes as well as combined measurements during validation, helping the team identify mounting or orientation problems that may be invisible in a single-axis configuration.

5.5 Controlled Frequency-Domain Support

The triggered FFT provides a second view of the vibration signal when the time-domain baseline indicates a possible change. This is useful for interpretation while avoiding a requirement to perform a full spectral calculation on every acquisition window. The repository also requires synthetic-tone checks and resource measurements before the FFT is considered a verified part of the final system [2].

5.6 Safety-Oriented Demonstration

The use of a guarded low-voltage motor or fan rig keeps the physical experiment bounded. Separate motor and ESP32 power reduces the risk of using the controller supply for the motor, and the requirement for a retained and guarded abnormal fixture limits the chance that a test object becomes a free-moving hazard [1].

5.7 Evidence Traceability

The repository goes beyond simple logging by requiring configuration identifiers, run identifiers, versioned parameters, immutable raw data and links between tests and final claims. This gives the project a traceability advantage: a future reader can distinguish the configuration

under which a result was produced instead of treating all observations as one undifferentiated dataset [2].

5.8 Educational Value

VibeGuard integrates embedded hardware, serial communication, signal processing, software versioning, experimental design and technical reporting. The project therefore provides a broad demonstration of engineering practice without requiring the team to make unsupported claims about industrial diagnosis.

5.9 Scope Discipline

A further advantage is the explicit separation of the semester MVP from future research. The repository states which features are out of scope, including cloud dependence, active cancellation, dense source attribution and industrial remaining-useful-life prediction. This makes it easier to assess whether a future change is an extension experiment or an unintended alteration of the original project claim [1].

CHAPTER 6
CHALLENGES

6.1 Mounting Sensitivity

Vibration measurement depends strongly on how the accelerometer is coupled to the test rig. A loose, flexible or inconsistent mount can change the measured signal independently of the machine condition. VibeGuard therefore requires a mount identifier and treats a material physical change as a configuration change [1, 2].

6.2 Acquisition Integrity

Real-time signal processing competes with acquisition for processor time and memory. A system that computes correct RMS values but silently loses samples cannot support a strong monitoring claim. The playbook consequently requires counted blocks, timing information and explicit checks for overruns and unexplained lost blocks [2].

6.3 Baseline Selection

A threshold is meaningful only relative to the operating condition used to establish the normal baseline. Motor speed, loading, mounting and environmental disturbance can change the vibration distribution. The project therefore requires multiple separated normal sessions and freezes the threshold only after pilot observations, instead of tuning it against the final held-out data.

6.4 Transient Disturbances

A short disturbance may cross a threshold without representing a sustained abnormal condition. VibeGuard addresses this by using persistence logic. The challenge is selecting a persistence definition that suppresses isolated disturbances without introducing excessive detection delay. The appropriate value has to be established during pilot testing and then frozen for final validation [2].

6.5 Frequency-Domain Resource Constraints

FFT processing requires memory and computation. Windowing, transform calculation and spectral-feature extraction must therefore be bounded so they do not interfere with the acquisition loop. The repository requires measurement of execution time, heap behaviour and dropped samples during FFT verification before the spectral stage is accepted [2].

6.6 Data Leakage and Correlation

Adjacent windows from one physical run are not independent observations. Random-window splitting can place near-identical portions of a single run into both training and test sets and can therefore overstate generalization. The VibeGuard protocol addresses this by keeping complete sessions together and holding out an entire day or remount for final evaluation [2].

6.7 Physical Safety

Generating a repeatable abnormal vibration must not create a dangerous rotating fixture. The project therefore requires guards, positive retention of any induced imbalance or disturbance mechanism, separate motor power, and power removal before physical adjustments.

These requirements constrain which abnormal conditions can be tested and are part of the technical methodology rather than administrative additions [1].

6.8 Configuration Drift

Small changes in firmware parameters, sampling settings, mounting or hardware revisions can alter the measured signal. Without configuration identifiers, the origin of a result becomes ambiguous. VibeGuard uses a combined configuration identifier linking hardware revision, mount, rig, firmware commit, parameter set and schema so that a change produces a visible evidence boundary [2].

6.9 Limited Physical Generalization

The semester demonstration uses a single controlled teaching rig. A successful result on this rig cannot automatically be generalized to industrial machines with different structures, loads, speeds, sensor locations or fault mechanisms. The repository explicitly limits the claim accordingly and does not support industrial fault-diagnosis or remaining-useful-life conclusions.

6.10 Incomplete Current Measurement Evidence

At the present project state, the repository describes the validation protocol and proposed acceptance criteria but does not contain the completed locked-test measurements needed to populate final performance statistics. Consequently, quantities such as final sensitivity, specificity, measured alert latency and false-alert rate must remain identified as not yet available rather than inferred from the design documents [1].

6.11 Schedule and Scope Management

The semester plan prioritizes stable acquisition, RMS-plus-persistence and the triggered FFT. The optional Mahalanobis comparator can be removed if it threatens the critical path or cannot be validated with sufficient independent data. This prioritization is necessary because adding an algorithm without completing the evidence chain does not improve the defensibility of the final report [2].

CHAPTER 7
CONCLUSION

VibeGuard defines a compact embedded approach for monitoring changes in vibration on a controlled low-voltage motor or fan rig. The current Level-1 architecture combines a rigidly mounted ADXL345 three-axis accelerometer, an ESP32-family controller, SPI acquisition, DC removal, RMS calculation, persistence logic, triggered Hann-windowed FFT support, local RGB indication and structured USB serial logging [1].

The project is deliberately narrower than a production predictive-maintenance system. Its central engineering question is whether a documented normal vibration baseline can be distinguished from a safe and deliberately induced abnormal condition on the selected teaching rig. The repository explicitly prevents the red state from being interpreted as a specific bearing or machinery diagnosis and keeps industrial remaining-useful-life prediction outside the semester claim.

The project also demonstrates that a technically complete monitoring system includes more than an algorithm. Sensor bring-up, acquisition integrity, mechanical mounting, safety, parameter freezing, configuration identifiers, data provenance and held-out validation are all part of the final evidence chain. The repository's execution playbook therefore treats failed tests, raw files, hashes, configuration records and validation outputs as first-class project artefacts [2].

At the current evidence state, the architecture and the validation procedure are documented, but final quantitative performance measurements are not available in the provided project sources. Proposed sensitivity, specificity, false-alert, latency and reproducibility criteria

must therefore remain clearly distinguished from achieved values. The final scientific conclusion should only be written after the frozen locked test has been completed and its independent results have been archived.

Taken together, the documented design provides a sound basis for a semester-scale demonstration of embedded vibration condition monitoring. Its main contribution is a traceable measurement-to-decision workflow whose claims can be expanded only when new evidence supports them.

CHAPTER 8
FUTURE SCOPE

8.1 Completion of Locked Validation

The immediate future scope is to complete the remaining project gates defined by the execution package. This includes confirming the as-built hardware and mounting arrangement, freezing firmware and parameters, completing regression checks, running the held-out evaluation and producing the final evidence package without post-result tuning [2].

8.2 Larger Independent Datasets

A larger collection of independent normal and abnormal sessions could improve confidence in the baseline parameters. Future data collection should continue to keep physical sessions together during validation so that increasing the dataset does not introduce leakage. Additional days, remounts and controlled operating variations could be used to characterize the stability of the learned baseline.

8.3 Improved Spectral Features

The triggered FFT can be extended with carefully selected frequency-band features once their usefulness and computational cost have been demonstrated. Such extensions should remain subordinate to acquisition integrity and should be frozen before final testing. More complex frequency-domain interpretation should not be presented as a diagnosis unless the corresponding evidence exists.

8.4 Optional Statistical Comparison

Mahalanobis distance or another multivariate anomaly score can be evaluated as an optional parallel comparator if sufficient independent normal data are available. The repository already treats this as optional and recommends removing it when it does not provide a clear benefit relative to schedule and validation risk [2].

8.5 Multi-Sensor Extensions

A later research version could investigate a second sensor or multiple measurement locations for spatial comparison. Any such extension should use a new configuration and a new validation plan so that the evidence for the one-sensor semester baseline remains unambiguous. The additional sensor could also support experiments on structural isolation and sensor-placement sensitivity.

8.6 Host-Side Visualization

Future versions could add a host-side dashboard that visualizes time-domain vibration, RMS trends, FFT bands and state transitions. Such a visualization layer would complement rather than replace the local decision path. The archived raw data and configuration identifiers should remain the authoritative evidence source.

8.7 Improved Mechanical Packaging

The physical platform could be developed with a more rigid enclosure, improved cable strain relief, a standardized accelerometer mount and a better documented abnormal-condition fixture. These improvements would reduce uncontrolled variation and make repeated demon-

strations easier while retaining the low-voltage safety boundary.

8.8 Broader Operating Conditions

A future research project could examine changes caused by different speeds, loads or controlled mechanical conditions. Such experiments would require new hypotheses and careful control of operating variables because a baseline learned under one operating condition should not automatically be treated as universal.

8.9 Toward a Research Platform

Longer-term research may investigate richer condition descriptors, multi-sensor fusion, automatic experiment manifests, uncertainty analysis or other ideas related to the deferred PIRG concept. The repository emphasizes that such work should be developed as a separate branch with its own hypothesis, safety review, data-independence plan and disclosure record rather than silently extending the semester VibeGuard claim [1].

8.10 Scope Preservation

The most important future-scope principle is to preserve the distinction between measured results and future hypotheses. New features should receive a new configuration identity and their own validation evidence. This keeps the original semester report reproducible while allowing the project to evolve into a more capable embedded condition-monitoring platform.

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